School of Engineering

REPUBLIC POLYTECHNIC

E211 – Operations Planning II

Quiz 1 Question for Lesson 9 E-Learning

Minimum Cost Flow

WonderToys has manufacturing centers located in Singapore, Bangkok and Manila. WonderToys is working closely with its regional partners and the plan is to ship the toys to customers in the other ASEAN countries – Jakarta, Phnom Penh, Naypyidaw, Hanoi, Kuala Lumpur, Vientiane except Bandar Seri Begawan. The latter is a transshipment node in its supply chain. Figure 1 shows the planned distribution network and Table 1 & Table 2 the respective supply and demand information.

How can WonderToys plan a distribution plan that meets the demand for its educational toys and minimize the total cost of shipment?

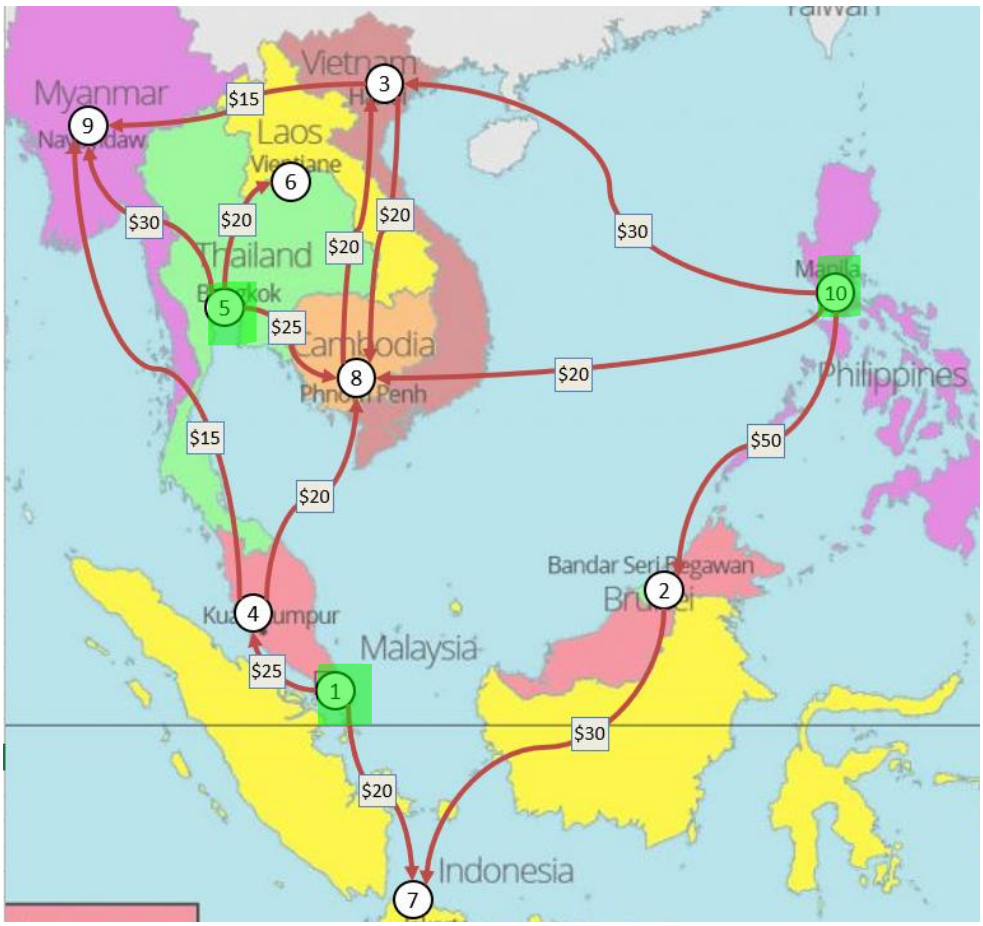


Figure 1: Toy Distribution Network

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	**1200**
[5] Bangkok	**800**
[10] Manila	**1000**
Total	**3000**

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	**0**
[3] Hanoi	**600**
[4] Kuala Lumpur	**800**
[6] Vientiane	**300**
[7] Jakarta	**600**
[8] Phnom Penh	**400**
[9] Naypyidaw	**300**
Total	**3000**

1) How do you define the decision variables used in the LP formulation? In total, how many decision variables are there?
   Answer: Let X_{ij} be the *number of units of toys (in '000s)* to be shipped from node i to node j. In total, there are 14 decision variables:
   X_{14}, X_{17}, X_{27}, X_{38}, X_{39}, X_{48}, X_{49}, X_{56}, X_{58}, X_{59}, X_{83}, X_{102}, X_{103}, X_{108}

2) Write down the objective function of the minimum cost flow problem using the decision variables defined earlier.
   Minimize total shipment cost, $Z = 25X_{14} + 20X_{17} + 30X_{27} + 20X_{38} + 15X_{39} + 20X_{48} + 15X_{49} + 20X_{56} + 25X_{58} + 30X_{59} + 20X_{83} + 50X_{102} + 30X_{103} + 20X_{108}$

3) What are the types of constraints in this model? ***Select all answers that apply.***
   Net flow constraints, non-negativity constraints, upper-bound constraints

4) Match the net flow constraint of each city with its formulation

Supply Quantities:

Manufacturing Plants	Supply Units
[1] Singapore	1200
[5] Bangkok	800
[10] Manila	1000
Total	3000

Demand Quantities:

Transshipment/ Demand Centers	Demand Units
[2] Bandar Seri Begawan	0
[3] Hanoi	600
[4] Kuala Lumpur	800
[6] Vientiane	300
[7] Jakarta	600
[8] Phnom Penh	400
[9] Naypyidaw	300
Total	3000

Supply Node 10: X102 + X103 + X108 = 1000	Net flow constraint for supply from Manila
Demand Node 7: - X17 - X27 = -600	Net flow constraint for demand from Jakarta
Demand Node 4: X48 + X49 -X14 = -800	Net flow constraint for demand from Kuala Lumpur
Transshipment Node 2: X27 - X102 = 0	Net flow constraint for transshipment city Bandar Seri Begawan
Supply Node 1: X14 + X17 = 1200	Net flow constraint for supply from Singapore

5) What is the upper-bound constraint in this network? *Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw*
   Arc 5 to 9: X_{59} **<=** 150

6) Formulate the non-negativity constraint.
   All X_{ij} >= 0

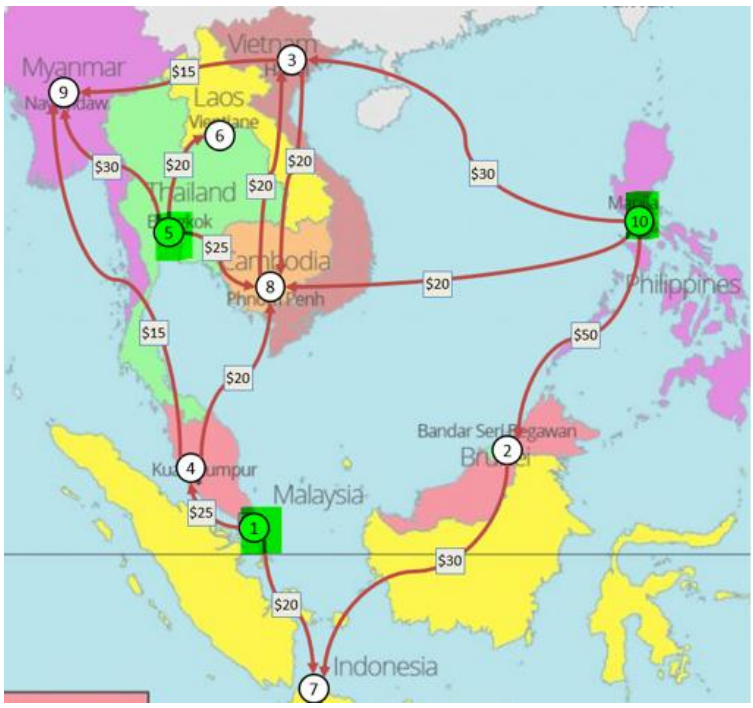
7) How would the constraints in your minimum cost flow LP model change if the plant in Singapore is able to supply 1500 units of toys instead of 1200? ***Select all answers that apply.***
   Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw.

Table 1: Supply Quantities

Manufacturing Plants	Supply Units
[1] Singapore	1500
[5] Bangkok	800
[10] Manila	1000
Total	3000

Table 2: Demand Quantities

Transhipment/ Demand Centres	Demand Units
[2] Bandar Seri Begawan	0
[3] Hanoi	600
[4] Kuala Lumpur	800
[6] Vientiane	300
[7] Jakarta	600
[8] Phnom Penh	400
[9] Naypyidaw	300
Total	3000



Supply Node 1: $X_{14} + X_{17} <= 1500$
Supply Node 5: $X_{56} + X_{58} + X_{59} <= 800$
Supply Node 10: $X_{102} + X_{103} + X_{108} <= 1000$

8) How would the constraints in your minimum cost flow LP model change if the demand at Hanoi has increased to 700? ***Select all answers that apply***.
Note: There is a maximum shipping capacity of 150 from Bangkok to Naypyidaw.

Demand Node 3: $X_{38} + X_{39} - X_{83} - X_{103} >= -700$
Demand Node 4: $X_{48} + X_{49} - X_{14} >= -800$

Demand Node 6: $-X_{56} >= -300$
Demand Node 7: $-X_{17} - X_{27} >= -600$
Demand Node 8: X_{83} (Flow out) $- X_{38} - X_{48} - X_{58} - X_{108}$ (Flow in) $>= -400$
Demand Node 9: $-X_{39} - X_{49} - X_{59} >= -300$

- [x] Demand Node 7: $-X_{17} - X_{27} >= -600$
- [x] Demand Node 3: $X_{38} + X_{39} - X_{83} - X_{103} >= -700$
- [ ] Demand Node 4: $X_{48} + X_{49} - X_{14} >= 800$
- [x] Demand Node 9: $-X_{39} - X_{49} - X_{59} >= -300$
- [x] Demand Node 4: $X_{48} + X_{49} - X_{14} >= -800$
- [x] Demand Node 6: $-X_{56} >= -300$
- [ ] Demand Node 9: $X_{39} + X_{49} + X_{59} = -300$
- [x] Demand Node 8: $X_{83} - X_{38} - X_{48} - X_{58} - X_{108} >= -400$